

1. CD4+ Th1 → IFN-γ

WHAT YOU SEE

Knight commander signaling 'IFN-γ' over the fortress wall

WHAT IT ENCODES

Containment of TB is a CD4+ Th1 cell-mediated response. Th1 cells secrete IFN-γ which activates macrophages to kill intracellular bacilli. This is the immunologic basis of both the granuloma and the IGRA/TST tests.

BOARD TRAP

It is NOT a humoral/antibody response — antibodies are useless against intracellular mycobacteria. Loss of CD4 (HIV) is what reactivates TB.

2. Activates macrophage

WHAT YOU SEE

Banner 'ACTIVATES MACROPHAGE' inside the wall

WHAT IT ENCODES

IFN-γ-activated macrophages fuse into epithelioid/giant cells forming the granuloma. Activated macrophages restrain but rarely sterilize bacilli, producing latent infection.

BOARD TRAP

TNF-α (not just IFN-γ) maintains the granuloma — anti-TNF biologics (infliximab) cause reactivation, so screen for LTBI before starting them.

3. Caseous necrosis

WHAT YOU SEE

'CASEOUS' center of the fortress courtyard

WHAT IT ENCODES

Caseating (cheese-like) granuloma is the histologic hallmark of TB. Central caseous necrosis surrounded by epithelioid macrophages, Langhans giant cells, and lymphocytes.

BOARD TRAP

NON-caseating granulomas suggest sarcoidosis (or Crohn). Caseation + acid-fast bacilli = TB.

4. Post-primary upper lobe cavitation

WHAT YOU SEE

'POST-PRIMARY UPPER LOBE CAVITATION' header, orange sky

WHAT IT ENCODES

Reactivation (post-primary) TB localizes to the high-oxygen apical/posterior upper lobes and cavitates. Cavitory disease = high bacillary load and high infectivity.

BOARD TRAP

Primary TB in adults often involves LOWER lobes without cavitation (Ghon focus); upper-lobe cavitation = reactivation.

5. Primary lower lobe — children

WHAT YOU SEE

'PRIMARY LOWER LOBE CHILDREN' blue panel

WHAT IT ENCODES

Primary infection seeds the mid/lower lung zones (Ghon focus) plus hilar nodes (Ghon/Ranke complex). Usually asymptomatic and self-limited in immunocompetent hosts.

BOARD TRAP

Don't expect cavities in primary TB — that is the reactivation pattern.

6. 10% lifetime reactivation risk

WHAT YOU SEE

Hourglass '18 MONTHS' and '10% LIFETIME RISK'

WHAT IT ENCODES

Immunocompetent latent TB carries ~10% LIFETIME reactivation risk, highest in the first 2 years. HIV co-infection raises this to ~10% PER YEAR.

BOARD TRAP

The 10% figure is lifetime for normal hosts; in HIV it becomes ~10%/year — a classic swapped-number trap.

7. Silicosis RR 30

WHAT YOU SEE

Miner with hard hat 'SILICOSIS RR 30'

WHAT IT ENCODES

Silicosis impairs macrophage function and is one of the strongest reactivation risk factors (relative risk ~30). Silica-laden macrophages cannot kill bacilli.

BOARD TRAP

Silicosis specifically predisposes to TB (and to upper-lobe progressive massive fibrosis).

8. HIV / CD4 <200 — highest risk

WHAT YOU SEE

Elderly figure 'CD4 <200', banner 'RR 21–30 HIGHEST RISK'

WHAT IT ENCODES

Advanced HIV (CD4 <200) is the top reactivation risk and causes atypical presentations: lower-lobe/disseminated/miliary disease, anergic TST. Start TB treatment, then ART.

BOARD TRAP

In HIV the TST may be falsely negative (anergy); use IGRA and a lower induration cutoff (≥5 mm).

9. Gastrectomy RR 2–5

WHAT YOU SEE

Stomach silhouette 'GASTRECTOMY RR 2–5'

WHAT IT ENCODES

Gastrectomy, chronic kidney disease/dialysis, diabetes, and malnutrition are moderate reactivation risk factors (RR 2–5).

BOARD TRAP

These are moderate (RR 2–5), not the extreme risks of HIV or silicosis — match the magnitude to the factor.

10. TST induration cutoffs

WHAT YOU SEE

Three banners: ≥ 5 mm, ≥ 10 mm, ≥ 15 mm

WHAT IT ENCODES

TST cutoffs depend on risk: ≥ 5 mm = HIV, recent contact, immunosuppressed/anti-TNF, fibrotic CXR; ≥ 10 mm = recent immigrants, IVDU, healthcare/prison, DM; ≥ 15 mm = no risk factors.

BOARD TRAP

A 6 mm reaction is POSITIVE in an HIV patient but NEGATIVE in a healthy low-risk person — interpret cutoff by risk group, not absolute size.

11. PPD reads at 48–72 hrs

WHAT YOU SEE

'TST / PPD — INDURATION 48–72 HRS'

WHAT IT ENCODES

TST is read at 48–72 hours measuring INDURATION (not erythema) transversely. It is a delayed-type (type IV) hypersensitivity reaction.

BOARD TRAP

Measure induration, NOT redness; reading before 48 h or after 72 h invalidates the test.

12. IGRA — IFN- γ release

WHAT YOU SEE

Tube 'IGRA = IFN- γ RELEASE', 'PREFER IGRA in BCG-vaccinated'

WHAT IT ENCODES

IGRA measures IFN- γ released by T cells exposed to TB-specific antigens (ESAT-6/CFP-10). One visit, objective, and not affected by BCG.

BOARD TRAP

Prefer IGRA over TST in BCG-vaccinated patients to avoid false positives; IGRA cannot distinguish latent from active TB.

13. BCG → false-positive TST

WHAT YOU SEE

'BCG' cannon ball, 'FALSE POSITIVE'

WHAT IT ENCODES

BCG vaccination can cause a false-positive TST. IGRA antigens are absent from BCG, so IGRA is 'NOT AFFECTED BY BCG'.

BOARD TRAP

A positive TST in a BCG-vaccinated immigrant may be BCG, not LTBI — confirm with IGRA before treating.

14. Screen first / no prior exposure

WHAT YOU SEE

Archer target 'RR 4–5 SCREEN FIRST', 'no prior exposure' figure

WHAT IT ENCODES

Before immunosuppression (anti-TNF, transplant, dialysis) screen for LTBI and treat to prevent reactivation. A truly unexposed host (negative IGRA/TST) needs no prophylaxis.

BOARD TRAP

Screening identifies latent infection to treat BEFORE biologics — don't start anti-TNF without LTBI screening.
